

Unit V:

Emerging trends in engineering management



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(4 Hrs.)

5.1 Participative management, conflict resolution, change management, quality management, innovation management and disruption

5.2 Recent engineering management concepts for managing ICT based projects and organizations

1. Participative Management

- A management style where employees are involved in decision-making processes.
- Improves motivation, job satisfaction, creativity, and ownership.
- Common in ICT projects using agile and team-based approaches.
- **Example:** *Developers participate in sprint planning and system design decisions.*

Process where subordinates share a significant degree of decision-making power with their immediate superiors.

-Stephen P. Robbins



Key characteristics

- Shared decision-making between managers and team members
- Encourages input from multiple organizational levels
- Builds commitment through involvement
- Particularly effective in knowledge-intensive environments

Potential benefits

- Increased employee engagement and morale,
- richer pool of ideas,
- greater buy-in to decisions,
- development of team members' skills.



Potential drawbacks

- Can be slower decision-making process,
- potential for disagreements and conflicts,
- may not be suitable for all situations (e.g., crisis situations requiring swift action).

2. Conflict Resolution

- The process of resolving disagreements between individuals or teams.
- Helps maintain teamwork, productivity, and project continuity.
- Effective conflict resolution is essential for project success.

Common conflict sources in engineering:

- *Technical approach disagreements*
- *Resource allocation disputes*
- *Schedule conflicts*
- *Scope changes and requirements ambiguity*
- *Personality clashes in team environments*





Conflict Management

- Conflict management is the process of identifying and handling conflicts in a fair, effective, and efficient manner.
- It involves **understanding the root causes, managing emotions, and promoting resolution** through communication and problem-solving strategies.
- Effective conflict management ensures **healthy work relationships, employee satisfaction, and organizational productivity.**



Role of Conflict management on HRM



- Improves employee morale and job satisfaction.
- Reduces turnover and absenteeism.
- Enhances teamwork and cooperation.
- Encourages constructive feedback and innovation.
- Strengthens leadership by fostering trust and communication.
- Poor conflict management results in stress, low productivity, and a toxic work environment

Human Behavior Factor	How It Can Cause Conflict
• Personality differences	• Clashes due to different styles
• Miscommunication	• Messages misunderstood
• Value/belief differences	• Disrespect or cultural tension
• Emotions like anger or stress	• Overreactions to small problems
• Power struggles	• Feeling ignored or controlled
• Competition for resources	• Fighting over money, time, support
• Unclear job roles	• Confusion, blaming each other
• Fear of change	• Resistance or frustration
• Group pressure	• “Us vs. them” mentality
• Poor communication	• Conflict due to bad speaking or listening

Types/Categories of Conflict

Based on Source/Origin

- Task Conflict
- Relationship Conflict
- Process Conflict

Based on Parties Involved

- *Intrapersonal Conflict*
- *Interpersonal Conflict*
- *Intragroup Conflict*
- *Intergroup Conflict*
- *Organizational Conflict*

Based on Issues/Content

- Substantive Conflict
- Emotional Conflict
- Cognitive Conflict

Based on Outcomes

- Functional Conflict
- Dysfunctional Conflict

Based on Visibility

- Manifest Conflict
- Latent Conflict



Conflict Management Modes/Methods

Five Primary Modes

1. Competing (Forcing)

- Assertive but uncooperative
- **Win-lose** outcome
- Use for urgent decisions, emergencies
- **Example:** *A manager enforces a decision on software selection despite team objections.*

2. Accommodating (Yielding)

- Unassertive but cooperative
- **Lose-win** outcome
- Use when relationship matters more
- **Example:** *A team member accepts a colleague's idea to avoid tension in the team.*

3. Avoiding (Withdrawing)

- Unassertive and uncooperative
- **Lose-lose** or no resolution
- Use for trivial issues, cooling-off
- **Example:** *A project manager delays discussion on a minor disagreement to focus on deadlines.*

4. Compromising (Sharing)

- Moderate assertiveness and cooperation
- **Partial win-win** outcome
- Use when equal power exists
- **Example:** *Two departments agree to split project resources equally.*

5. Collaborating (Problem-Solving)

- Assertive and cooperative
- **Win-win** outcome
- Use for important relationships
- **Example:** *Developers and testers jointly redesign a workflow to improve quality and speed.*

Alternative Methods

- **Mediation**

- Neutral third party facilitates
- Parties control outcome
- Voluntary and confidential
- Example: *HR mediator helping resolve workplace harassment complaint*

- **Arbitration**

- Third party makes binding decision
- More formal process
- Legally enforceable
- Example: *Labor union dispute resolved by independent arbitrator*

Organizational Change

Meaning/Concept

- *Change is status quo(Resist to opposite change)*
- *Very essential for continuous survival and adopt with the new changing environment*
- process of **changing** an **organization's** strategies, processes, procedures, technologies, and culture

Nature of Change

- *Change is natural and inevitable*
- *Change affects the working process, procedure and system*
- *Affected by environment*
- *Essential for survival*
- *Need planned change*

Change Management

- *Change Management is a **systematic and structured approach** to dealing with transformation in an organization by **planning, implementing, controlling, and sustaining change**.*
- *In engineering and ICT organizations, change may involve **new technologies, systems, processes, organizational structures, or project requirements**.*



For computer engineers, change management mainly deals with:

- *Software updates and version changes*
- *System upgrades and migrations*
- *Technology adoption (AI, cloud, DevOps, cybersecurity tools)*
- *Requirement and scope changes in ICT projects*

Change Management Models

- Change management models provide **structured frameworks** to plan, implement, and control changes in **technology, systems, processes, and people** in ICT projects.

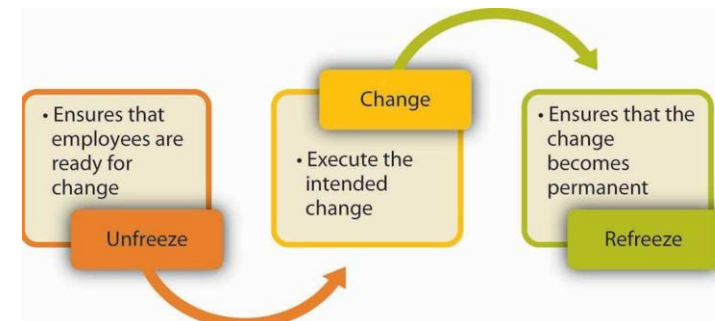
Lewin's Change Management Model

- Proposed by **Kurt Lewin** in 1947.
- Helps organizations **move from an existing state to a desired new state** smoothly.
- Focuses on **reducing resistance, implementing change, and stabilizing improvements.**
- this is a **simple and widely used three-stage model** suitable for managing **technological and system changes.**
- A **three-stage model** used to manage change in systems, technology, and processes.

1. Unfreeze

2. Change

3. Refreeze



- **Unfreeze**

- Recognizing the need for technological change
- Preparing systems and computer engineers for change
- Reducing resistance to new technology

Example: *Informing engineers about shifting from local servers to cloud systems*

- **Change**

- Implementing new technology or system
- Training engineers to use new tools
- Adjusting workflows and processes

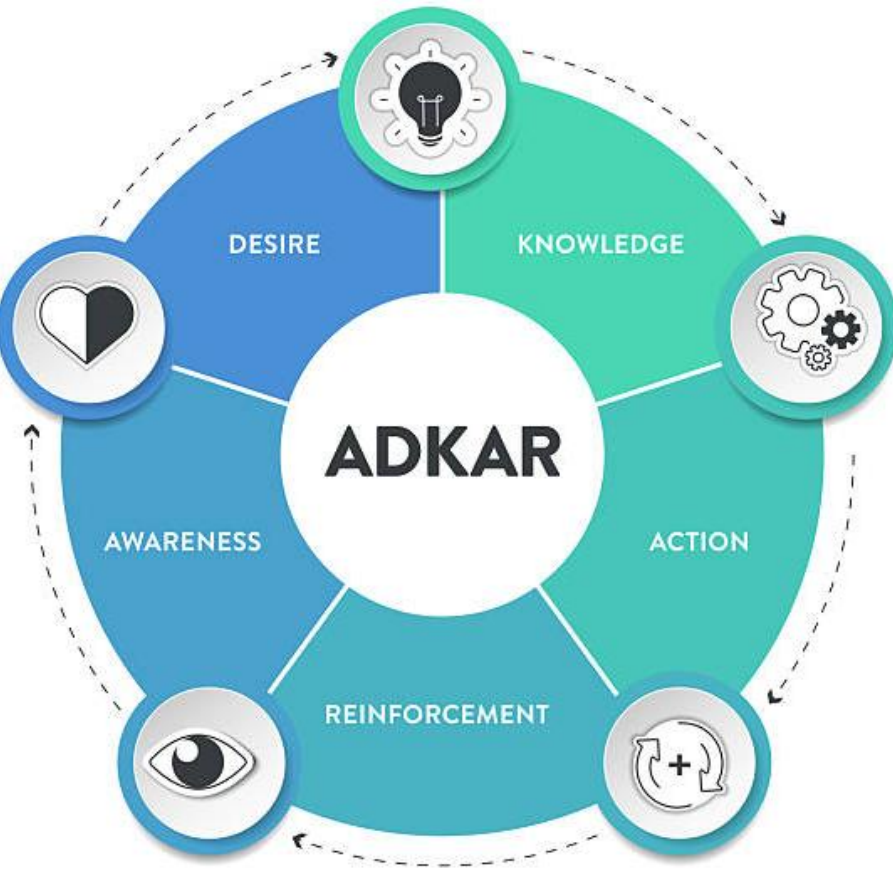
Example: *Migrating applications to a cloud platform*

- **Refreeze**

- Stabilizing the new system
- Documenting procedures and standards
- Making the change permanent

Example: *Making cloud deployment the standard practice and disabling old servers*

ADKAR Model



- Propounded by Jeff Hiatt (1998)
- Focuses on individual change through five stages: **Awareness, Desire, Knowledge, Ability, Reinforcement**
- Helps organizations implement and sustain change effectively

- **Awareness (A):**

- Understanding the need for change

Example: *Engineers are informed that old servers are causing slow application performance*

- **Desire (D):**

- Willingness to support and participate in the change

Example: *Engineers motivated to adopt **cloud computing** for faster, scalable systems*

- **Knowledge (K):**

- Learning how to change and what to do

Example: *Training on **cloud deployment, virtualization, and monitoring tools***

- **Ability (A):**

- Applying the knowledge to implement the change

Example: *Engineers successfully migrate applications to the cloud environment*

- **Reinforcement (R):**

- Ensuring the change sticks and is sustained

Example: *Regular audits and updates ensure **cloud systems remain the standard**, old servers are decommissioned*

4. Quality Management

- Quality management ensures that ICT products and services **meet required standards and user expectations.**
- Includes quality planning, assurance, and control
- Focuses on performance, reliability, security, and usability
- Uses testing, audits, and international standards
- **Example:** *Software testing and code reviews to ensure system reliability.*

5. Innovation Management

- Innovation management involves **planning and controlling the introduction of new ideas and technologies.**
- Promotes continuous improvement and competitive advantage
- Encourages R&D and creative problem-solving
- Essential in fast-changing ICT environments

Example: *Implementing AI-based automation in business systems.*



6. Disruption Management

- Disruption management deals with **radical technological changes** that can replace existing systems.
- Requires flexibility and strategic adaptation
- Helps organizations respond to emerging technologies
- Prevents business and system obsolescence



Example: *Traditional IT firms adapting to SaaS and cloud platforms.*

5.2 Recent Engineering Management Concepts for ICT Projects & Organizations

- Modern ICT-based organizations face **rapid technological shifts**.
- Recent engineering management concepts emphasize **flexibility, digital integration, and resilience**:
- **Agile Project Management**
 - Iterative development, customer collaboration, adaptability.
 - Widely used in software and ICT projects to handle uncertainty.
- **DevOps Practices**
 - Integration of development and operations for faster delivery.
 - Encourages automation, continuous integration, and deployment.
- **Digital Transformation Leadership**
 - Managing adoption of cloud computing, IoT, and AI.
 - Focus on aligning technology with business strategy.
- **AI & Data-Driven Decision Making**
 - Using predictive analytics and machine learning for project planning.
 - Enhances risk management and resource allocation.
- **Cybersecurity Management**
 - Embedding security into project lifecycles.
 - Essential for ICT organizations handling sensitive data.
- **Sustainability & Green ICT**
 - Managing energy-efficient systems and eco-friendly practices.
 - Increasingly important for compliance and corporate responsibility.
- **Remote & Hybrid Work Management**
 - Tools and strategies for distributed teams.
 - Requires strong digital collaboration platforms.